Samet Atak

ITU Control and Automation Engineering
Autonomous Systems & Robotics

GitHub · LinkedIn · sametatakwork@gmail.com

sametatak.github.io

PROJECTS

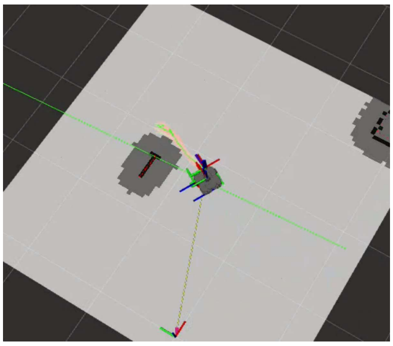
MPPI-Based Local Trajectory Planner for UGV

Graduation Project - Advanced Control & Navigation

Developing a high-performance local trajectory planner for autonomous ground vehicles (UGVs) based on Model Predictive Path Integral (MPPI) control within the ROS ecosystem:

- **Dynamic Obstacle Avoidance:** Implemented real-time reactive navigation capable of safely avoiding non-static obstacles in highly dynamic environments.
- **Optimal Control:** Utilized parallelized sampling-based predictive control loops to optimize non-linear vehicle dynamics and trajectory tracking.
- **ROS Integration:** Designed clean, highly responsive control nodes integrated with existing localization and mapping pipelines.

▷ **Project Repository:** **GitHub Link**



MPPI Local Planner Simulation in ROS

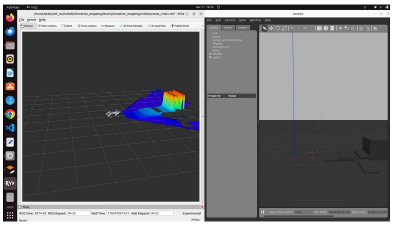
Quadruped Robot Control & Elevation Mapping

Simulation and Control Architecture

Developed locomotion and environmental perception capabilities for a quadruped robot in Gazebo:

- **Legged Control:** Implemented robust leg control algorithms for trot walking and stair climbing maneuvers.
- **Elevation Mapping:** Successfully integrated ETH Zurich's elevation mapping package into the ROS/Gazebo simulation environment.
- **Perception:** Generated real-time 2.5D terrain maps using simulated sensor data (visualized in RViz) to enhance the robot's spatial awareness.

▷ **Project Repository:** **GitHub Link**



Elevation Mapping in RViz & Gazebo

Autonomous Robot Developer

TÜBİTAK Supported R&D Project

End-to-end development of autonomous navigation and perception systems:

- **Embedded Software:** MCU-based controller software development on STM32 platforms.
- **Navigation:** Path planning, obstacle avoidance, and dynamic mapping within the ROS environment.
- **Sensor Fusion:** Integration of Intel RealSense visual odometry, GPS, and IMU data.
- **Autonomy:** Successfully achieved full autonomous waypoint navigation under real-world field conditions.

▷ **Project Details: Visit Website**

Autonomous Robot Platform

ACADEMIC PUBLICATIONS & TECHNICAL REPORTS

- **Technical Report:** *Advancing Ensemble Regression: Automated Hyperparameter Tuning of the Diversity-Accuracy Trade-Off in Sign Diversity Metric*
  University of Kassel, Germany (2025) – Academic research on optimizing the balance between diversity and accuracy in ensemble regression models.
  [View Report]

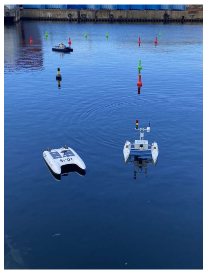
- **Article:** *Real-Time Face Recognition and Tracking in Matlab*
  IEEE Formatted Project Paper (2025) – Design of high-efficiency algorithms for real-time face detection and tracking in video streams.
  [View Article]

ACHIEVEMENTS & AWARDS

NJORD Autonomous Ship Challenge - World Champion (2023)

ITU Autobee ROS Team

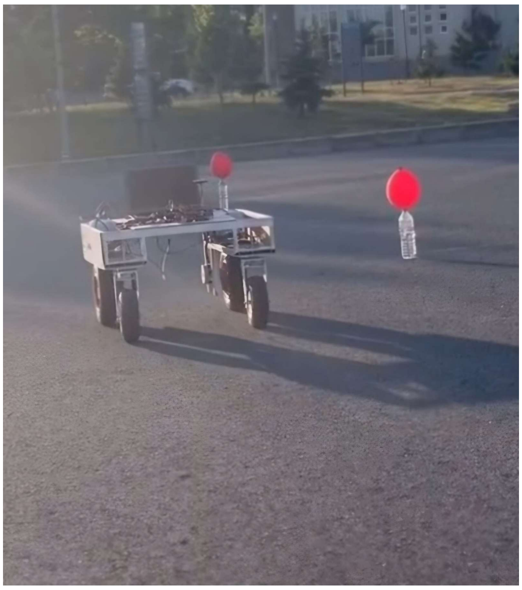
In the international autonomous marine systems competition held in Norway, our team secured **1st Place globally**, outperforming prestigious technical universities such as **MIT (Massachusetts Institute of Technology)** and various global competitors. Our success was driven by a high-performance software architecture that executed complex navigation, dynamic obstacle avoidance, and precision autonomous docking tasks.



NJORD World Championship - Norway

Teknofest TIKA Competition - Finalist
ITU Autobee Team

Successfully completed all autonomous mission scenarios in challenging environments requiring innovative autonomous solutions and advanced system integration.



Teknofest TIKA Team Photo